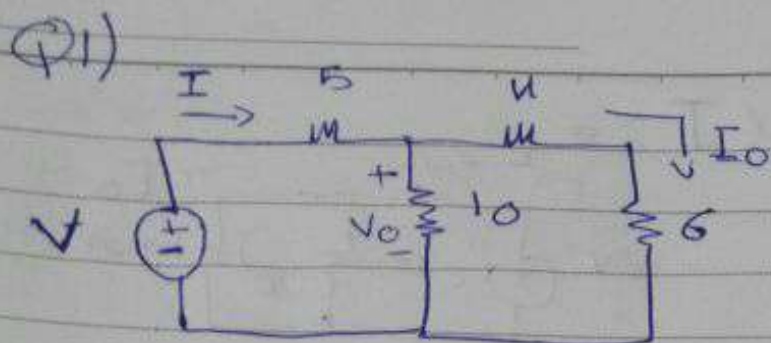


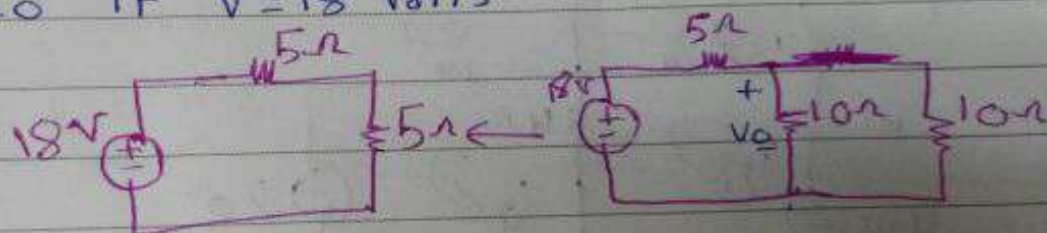


* Find R_{eq} equivalent

$$R_{eq} = [(4+6) \parallel (10)] + 5$$

$$= 10 \Omega$$

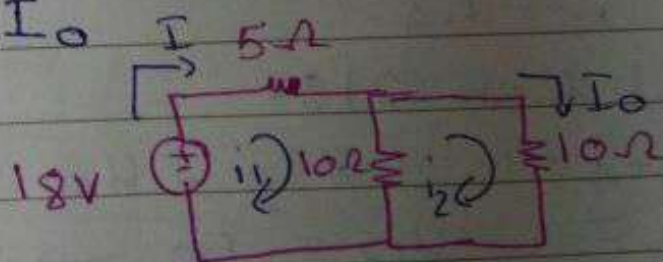
* Find V_0 IF $V = 18$ volts



$$V_{R5} = V_0 = \frac{18 \times 5}{5+5} = 9V$$

IF $I = -9A$ Find I_0

$$i_1 = I = -9A$$



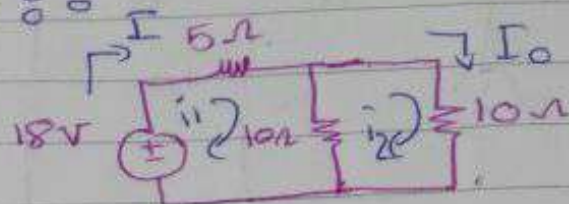
at Mesh (2) $\circ = 0$

$$-10 i_1 + (10+10) i_2 = 0$$

$$20 i_2 = 9 \Rightarrow i_2 = I_0 = -4.5A$$

~~* IF $I_0 = -4.5A$~~

* IF $I_0 = -4A$ Find $I = ?$



$$i_2 = I_0 = -4A$$

at Mesh (2) $\circ$

~~$$15i_1 - 5i_2 = 18$$~~

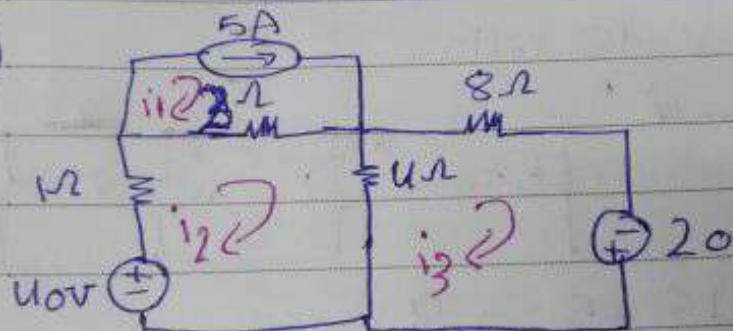
~~$$15i_1 - 22 = 18$$~~

$$-10i_1 + 20i_2 = 0$$

$$10i_1 = 8$$

$$i_1 = I = -8A$$

Q2)



* Find The current on 1Ω and 8Ω $\circ$

$$i_1 = 5A$$

at Mesh (2) $\circ$

$$-2 \times i_1 + 7i_2 - 4i_3 = 40$$

$$7i_2 - 4i_3 = 50 \rightarrow [1]$$

at Mesh (3) $\circ$

$$-4i_2 + 12i_3 = 20 \rightarrow [2]$$

$$i_2 = 10A$$

$$i_3 = 5A$$

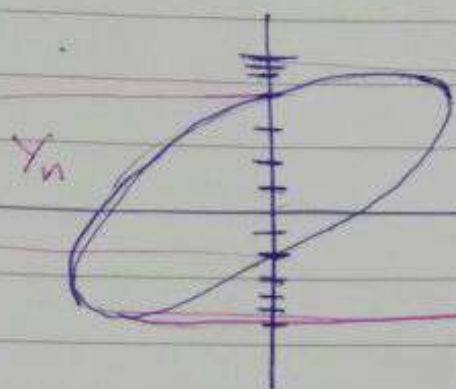
$$I(1\Omega) = i_2$$

$$I(8\Omega) = i_3 = 5A$$

* Find The power on (4Ω) $\circ$

$$P = (i_2 - i_3)^2 \times 4 = 100W$$

Q 5)

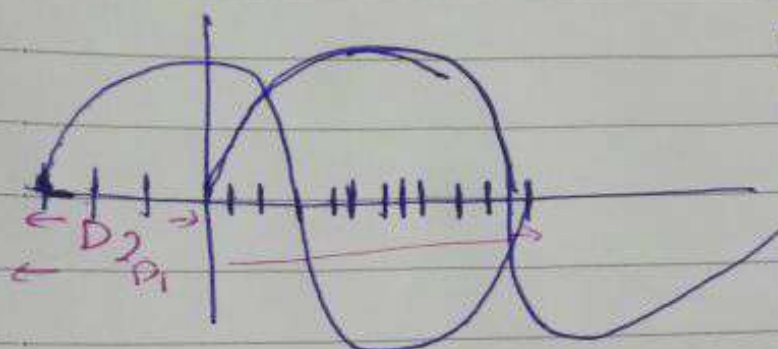


$$Y_n = 6$$

$$Y_m = 12$$

$$\sin^{-1}\left(\frac{Y_n}{Y_m}\right) = 30^\circ$$

Q 6 :: 0



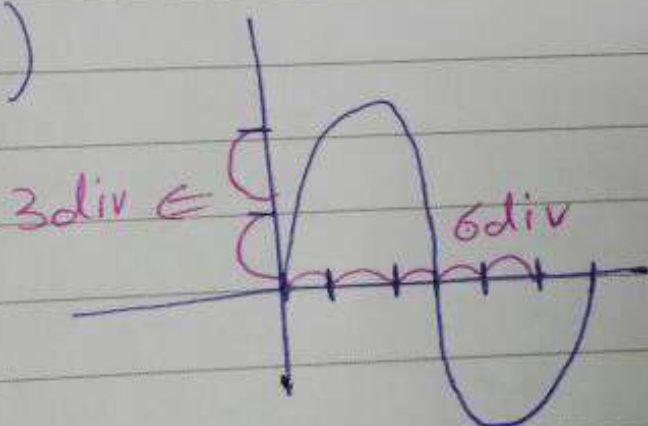
$$D_2 = 3$$

$$D_1 = 14$$

$$\Theta = \left(\frac{D_2}{D_1}\right) 360^\circ$$

$$\Theta = 77.14^\circ$$

Q 7)



Vertical Sensitivity = 2 V/div

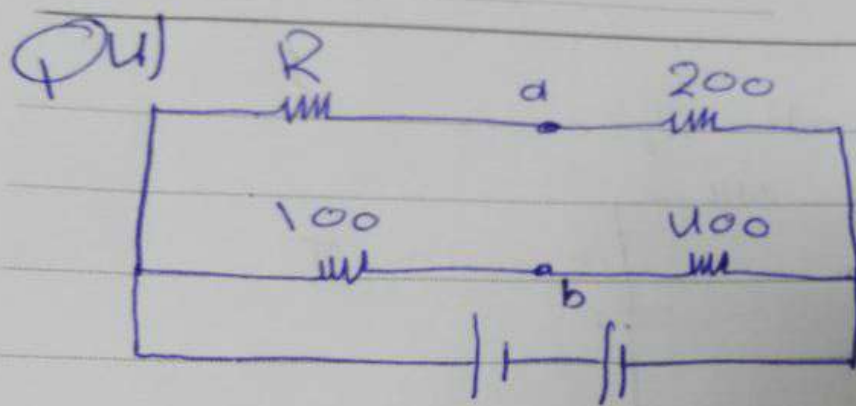
Horizontal Sensitivity = 5 ms/div

$$V_p = (6V) \rightarrow 3 \text{ div} \times 2 \frac{V}{\text{div}}$$

$$\text{The Time} = (3ms) \rightarrow 6 \text{ div} \times 5 \frac{ms}{\text{div}}$$

$$\text{Frequency} = \frac{1}{3ms} = 333.33 \text{ Hz}$$

$$RMS = \frac{1}{\sqrt{2}} \times 6 = 4.24 \text{ V}$$

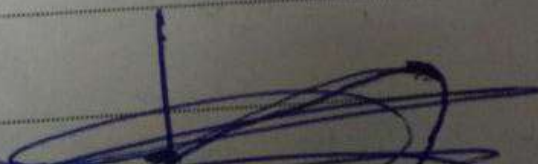


If $V_{ab} = 0$ Find R Ω

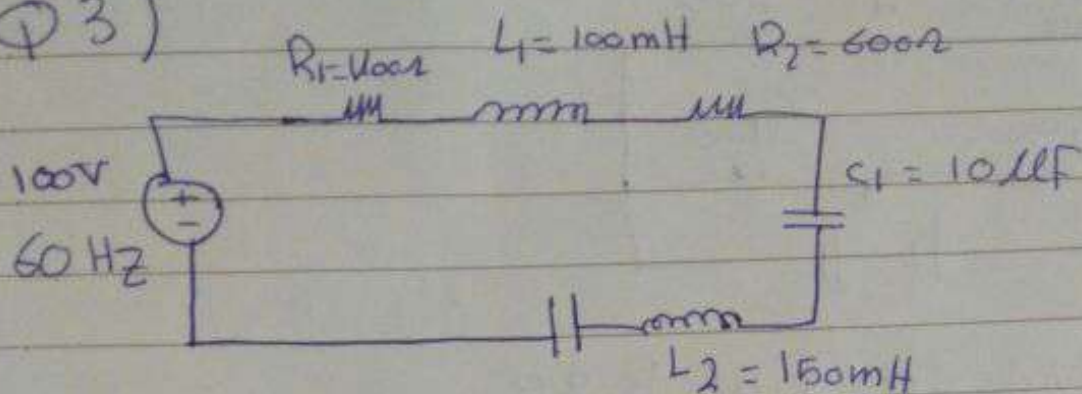
$$400 \times R = 100 \times 200$$

$$R = 50 \Omega$$

~~Q5~~



Q3)



$$X_{L1} = 2\pi f L_1 j = 37.699 j$$

$$X_{L2} = 2\pi f L_2 j = 56.5486 j$$

$$X_{C1} = \frac{1}{2\pi f C_1} = -265.258 j$$

$$X_{C2} = \frac{1}{2\pi f C_2} = -132.629 j$$

* Find The Impedance $\circ^\circ$

$$Z = R_1 + R_2 + X_{L1} + X_{L2} + X_{C1} + X_{C2}$$

$$Z = 1045 \Omega$$

* Find The angle $\circ^\circ -16.89^\circ$

* Find The Power Factor $\circ^\circ$

$$\cos(-16.89) = 0.95 \text{ Leading}$$